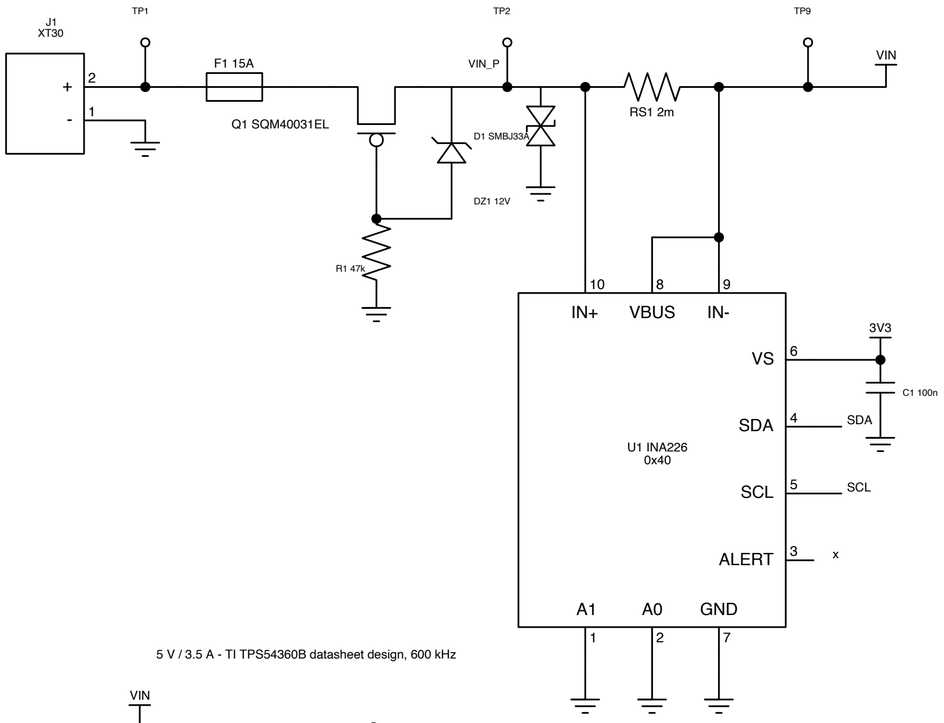
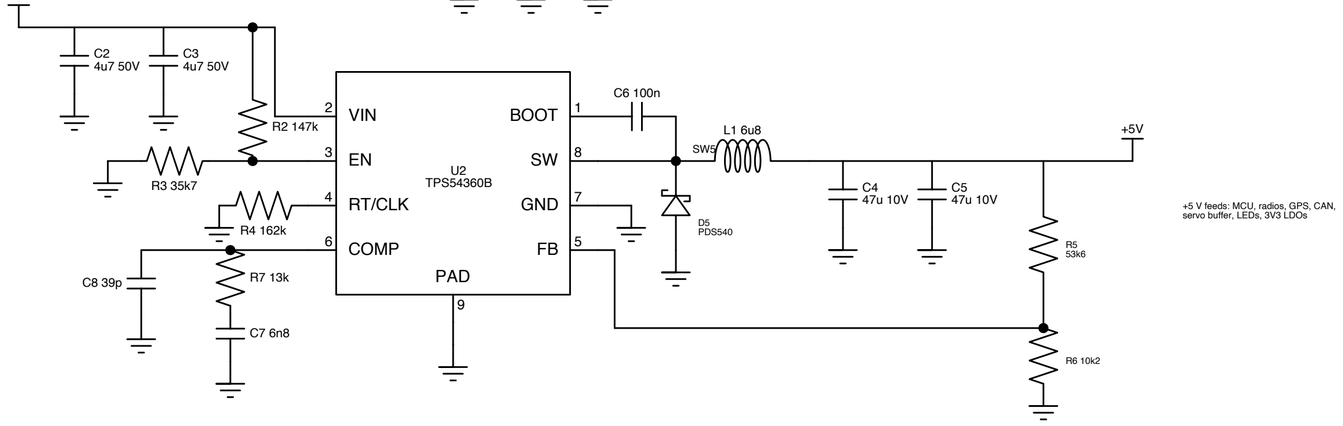
BATTERY INPUT - 2S-6S (6.5-26 V), 15 A fused, reverse-polarity protected, V / I monitored

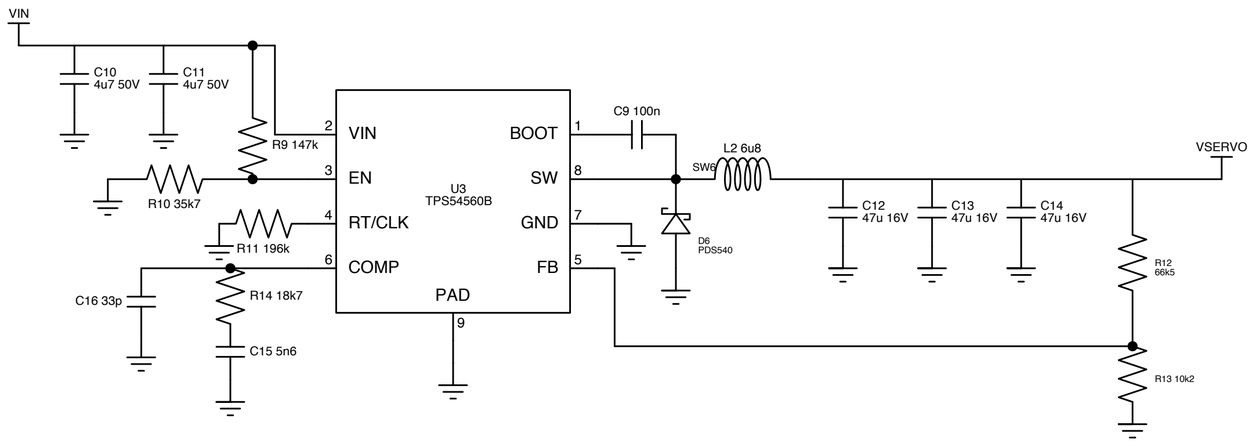



5 V / 3.5 A - TI TPS54360B datasheet design, 600 kHz

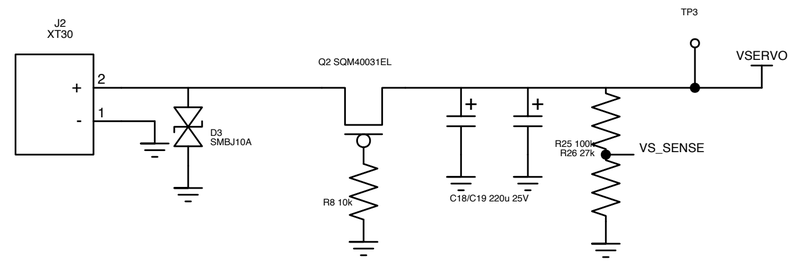



+5 V feeds: MCU, radios, GPS, CAN, servo buffer, LEDs, 3V3 LDOs

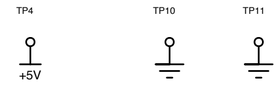
6.0 V / 5 A servo rail (Mode A) - TI TPS54560B, 500 kHz




SERVO BEC INLET (Mode B: 12 A, up to 8.4 V)




RAIL TEST POINTS

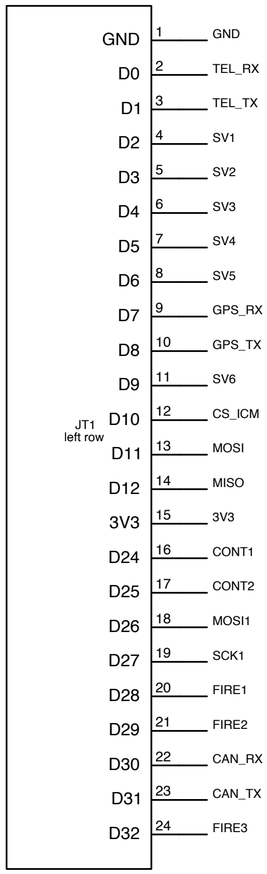


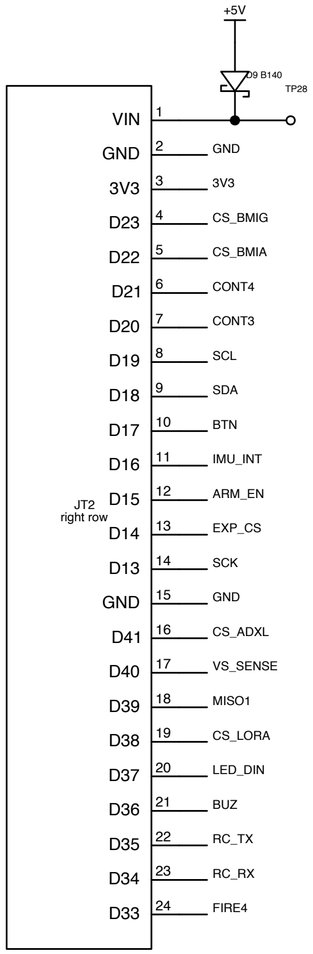

NOTES

1. Q1 / Q2 conduct only with the battery the right way round (body diode, then the gate pulled low); DZ1 holds VGS inside +-20 V on 6S + transients.
2. RS1 -> INA226: 81.92 mV full scale = 41 A; Kelvin lines to the shunt pads.
3. Mode A: U3 fitted, nothing on J2. Mode B: leave L2 off, BEC on J2.
4. Both bucks: UVLO on at 6.0 V. The 5 V rail is sized for RFD900x bursts (1 A).
5. 6 V compensation from TI's equations (fco 28 kHz, 85 uF derated COUT).

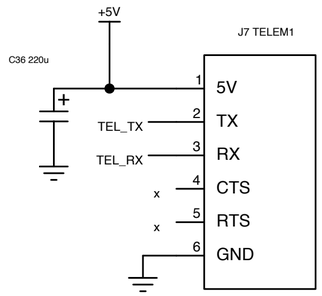
| | |
|---|---|
| POWER | |
| TWILIGHT-FC REV A (cad_bench/projects/twilight_fc.py) | |
| checked against the layout netlist | |
| SHEET 1 / 4 | REV A |
| DATE 2026-11-05 | SIZE A3 |

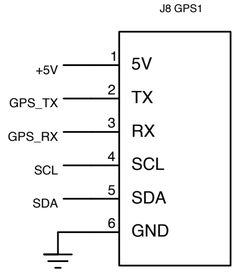
TEENSY 4.1 (i.MX RT1062, 600 MHz) - two 1x24 sockets; every edge pin used

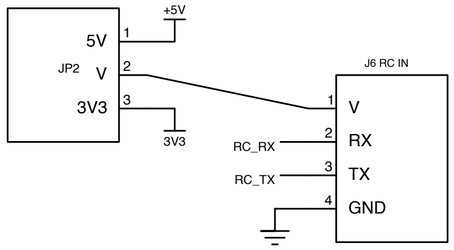


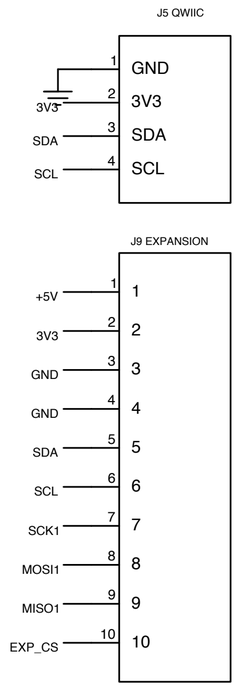




CONNECTORS

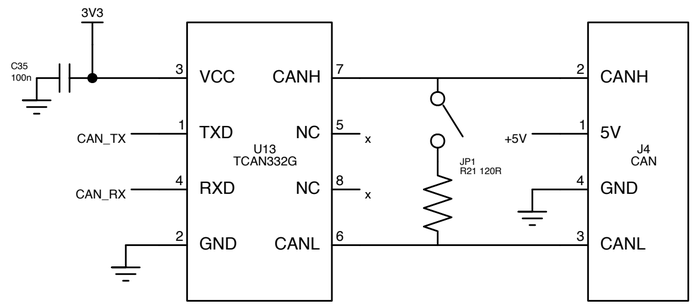


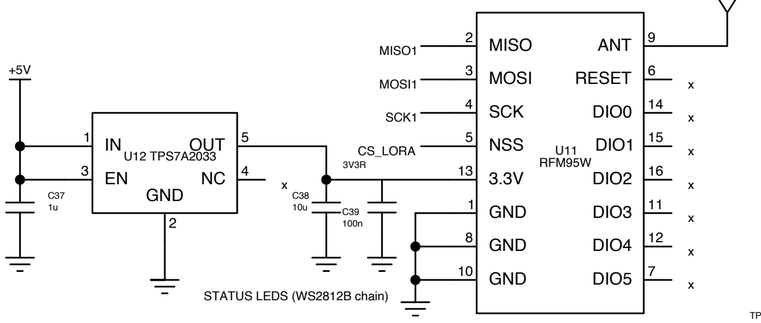



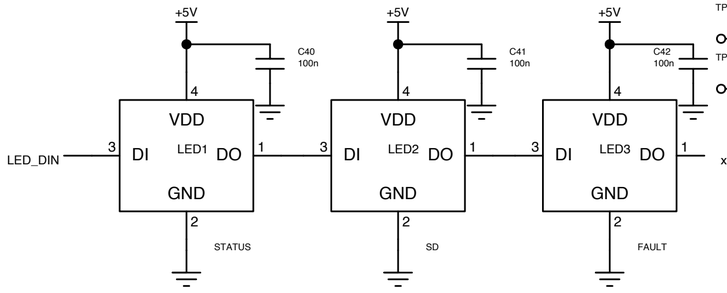

CAN-FD (CAN3)



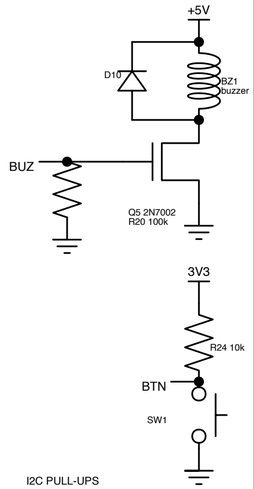

LoRa BEACON - RFM95W 915 MHz (SPI1)




STATUS LEDS (WS2812B chain)




UI

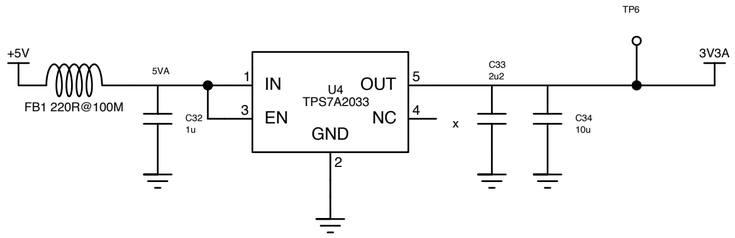


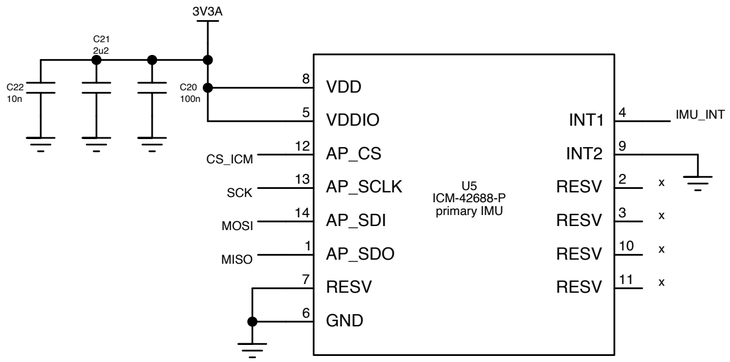

| | |
|---|---|
| COMPUTE + I/O | |
| TWILIGHT-FC REV A (cad_bench/projects/twilight_fc.py) | |
| Teensy 4.1 pin numbers in the boxes | |
| SHEET 2 / 4 | REV A |
| DATE 2026-11-05 | SIZE A3 |

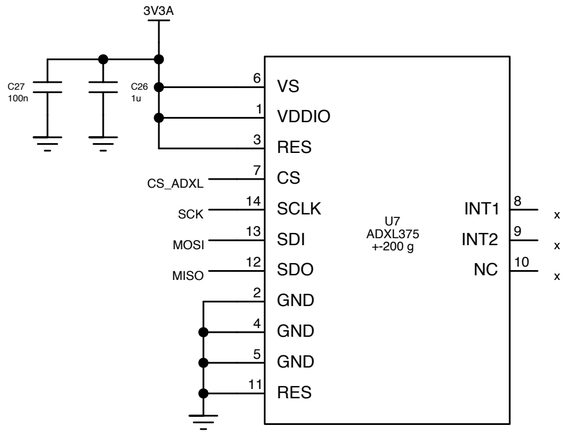
ANALOG 3.3 V - ferrite + its own LDO, at the sensor island

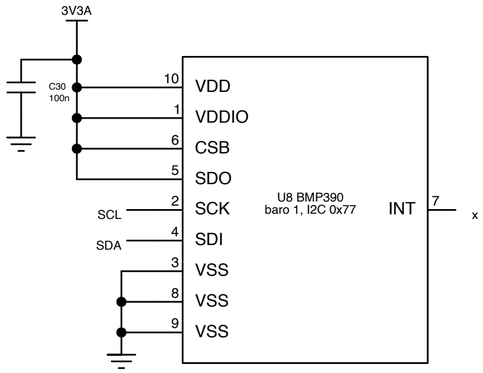
Sensor datum: U5 (ICM-42688-P) centre, board X 130.0 Y 25.0 mm (silkscreened).

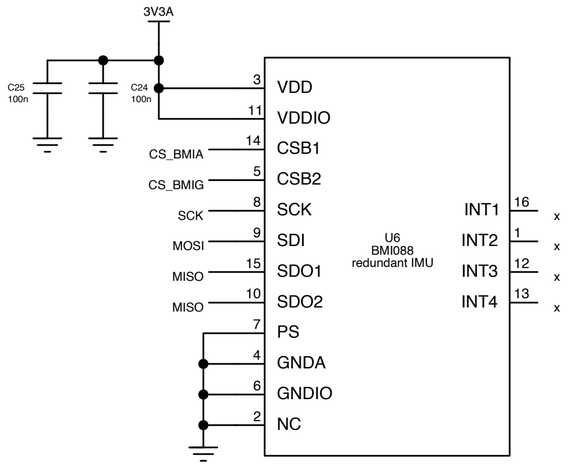
Every sensor on 3V3A; SPI0 at 24 MHz with one chip select each; I2C at 400 kHz.

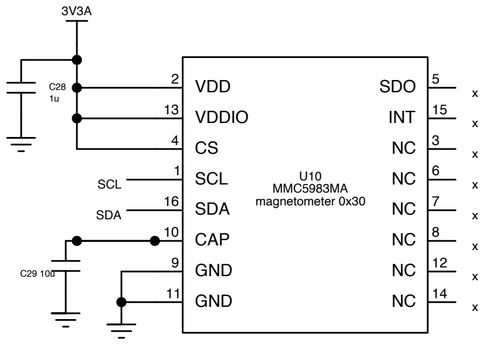


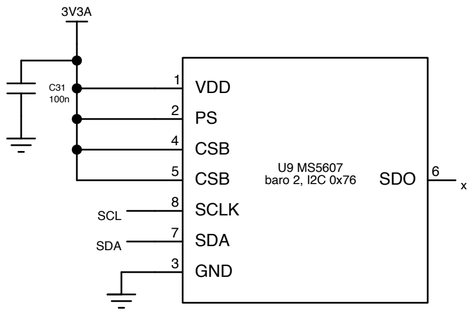



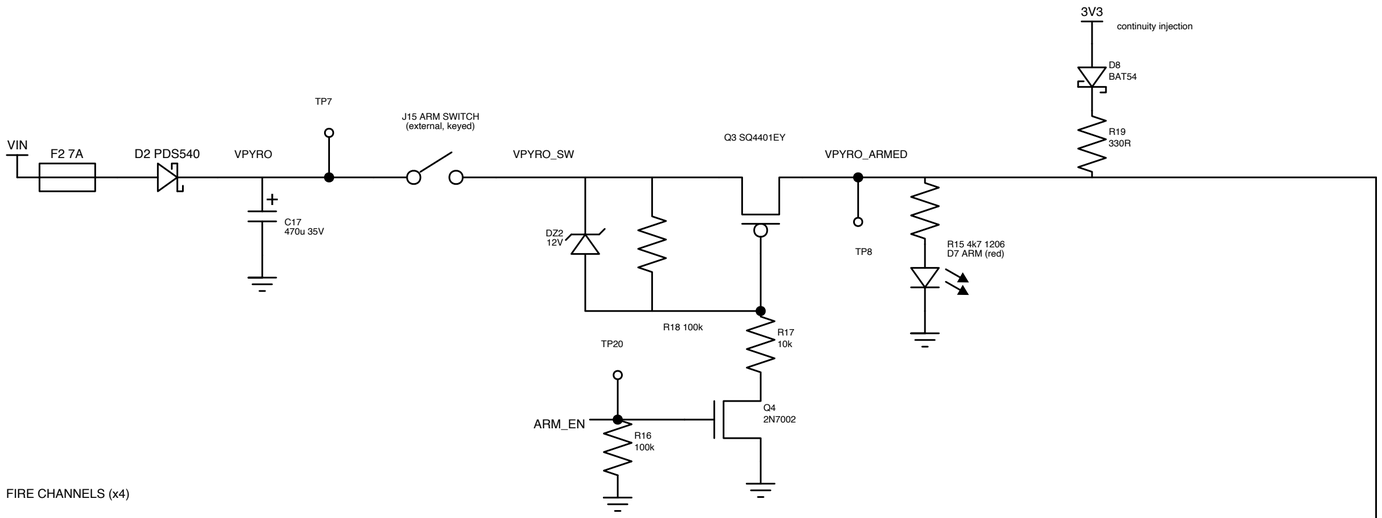



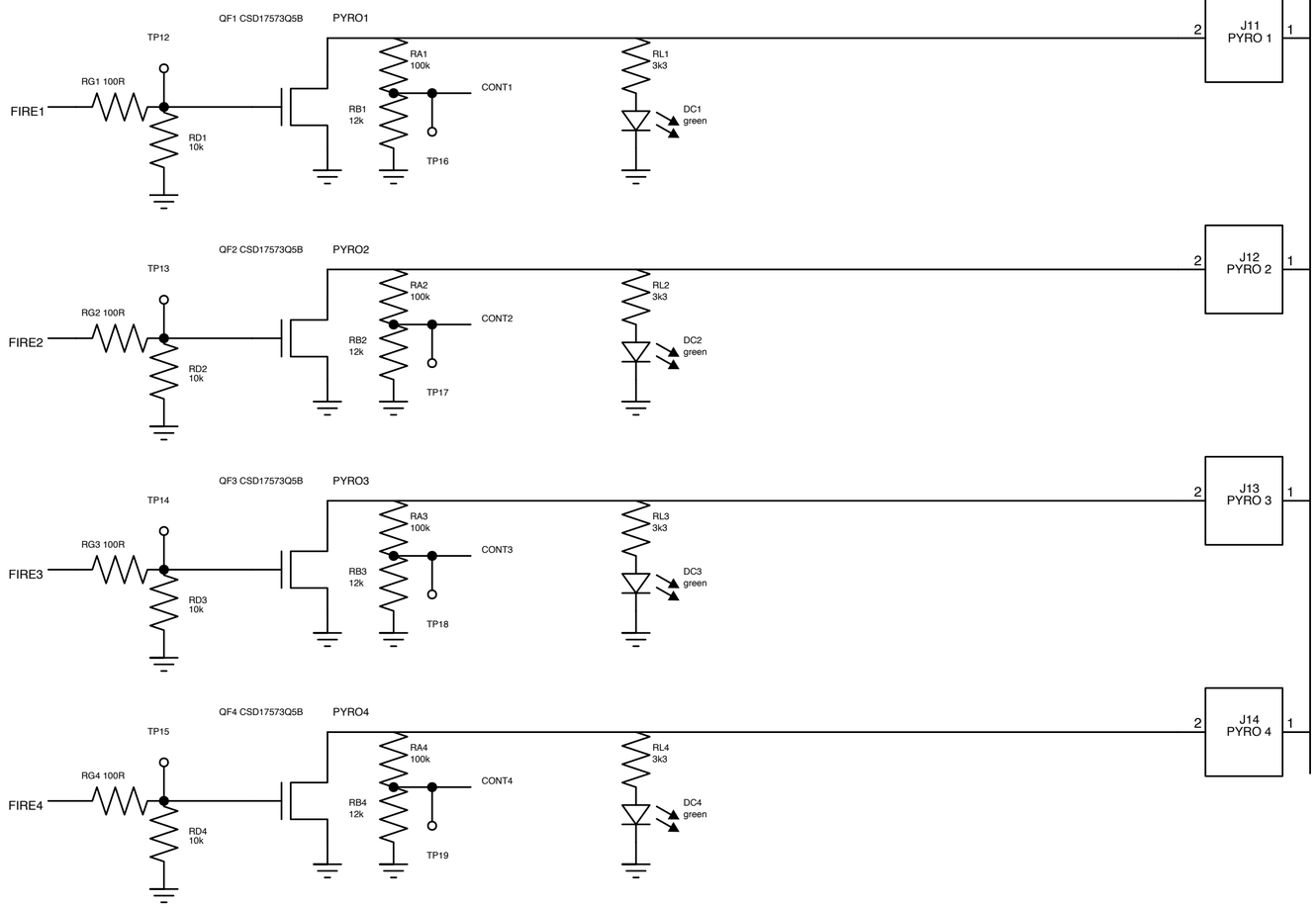




| | |
|---|---|
| SENSORS | |
| TWILIGHT-FC REV A (cad_bench/projects/twilight_fc.py) | |
| SPI0 inertial bus; I2C baros + magnetometer | |
| SHEET 3 / 4 | REV A |
| DATE 2026-11-05 | SIZE A3 |

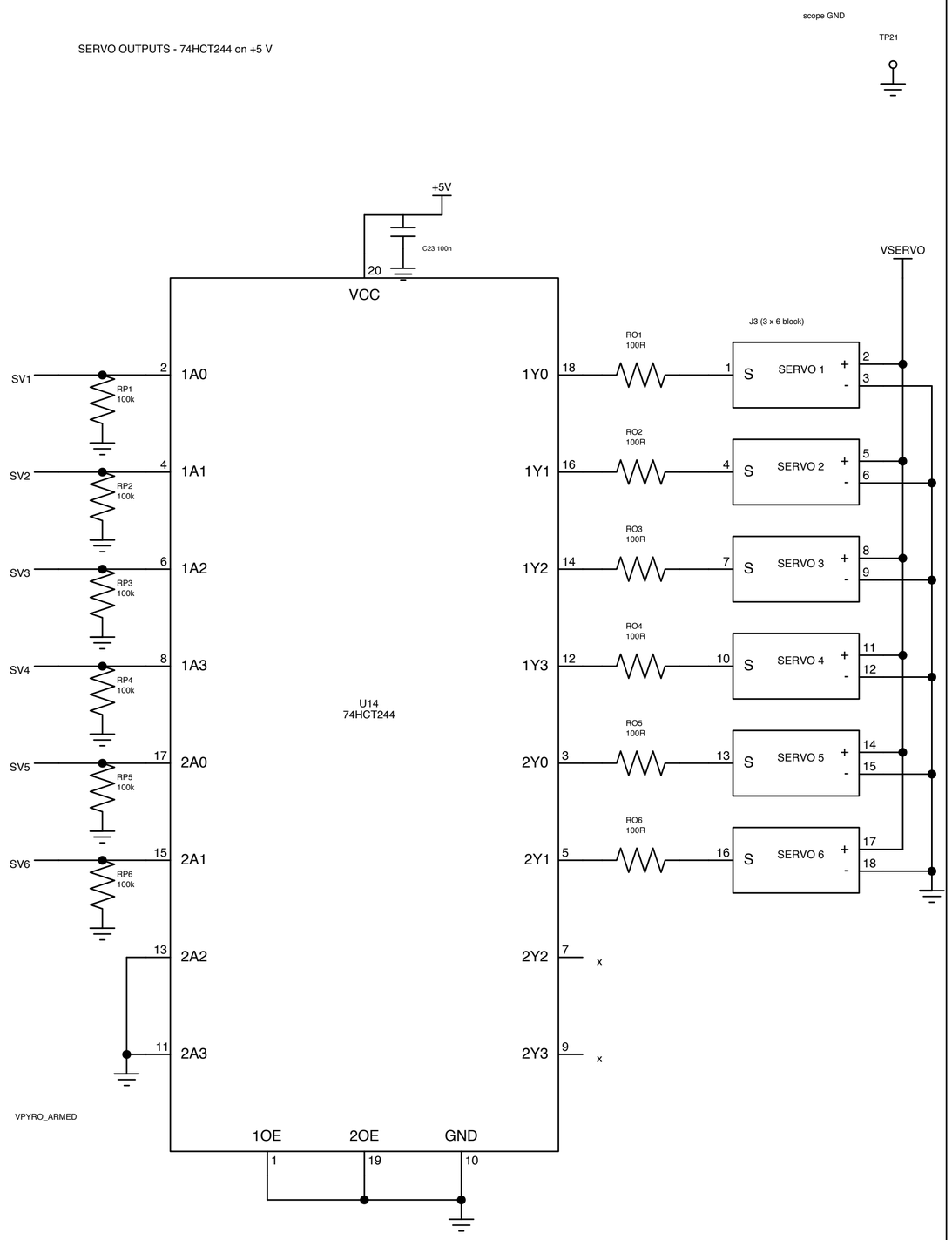
PYRO SUPPLY + ARMING - current flows only with the keyed switch closed AND ARM_EN high AND the channel's gate driven




FIRE CHANNELS (x4)




SERVO OUTPUTS - 74HCT244 on +5 V




PYRO + SERVO

TWILIGHT-FC REV A (cad_bench/projects/twilight_fc.py)

pyro: switch AND ARM_EN AND gate

| | |
|---|---|
| SHEET 4 / 4 | REV A |
| DATE 2026-11-05 | SIZE A3 |